Analysis of Algorithms (BDSF25M/A)

Date: *25-Feb-2026* **Quiz 01** **Time:** *20 min*

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Honor Code: By signing below, I am declaring under oath that I will solve the quiz by myself solely and will not Take any help from neighboring fellows.

Signature

Grading	CLO-1	CLO-2	CLO-3	CLO-4
	/	/	/	/

1. Define Algorithm.

An Algorithm is a well defined computational procedure that take value as input the produces it and returns / transforms into output. It acts as a tool for problem to transform input into output.

2. What are the properties of an algorithm.

An Algorithm must satisfy following properties to be valid:

→ Input: It must be provided with some value from outside.
→ Output: It must produce atleast one result.
→ Definiteness: Every instruction must be clear (unambigious)
→ Finiteness: procedure must terminate after countable no of steps.
→ Effectiveness: each step must be simple enough to be performed manually with a pen and paper.

3. What is the meaning of following: **Algorithms is technology?**

The overall performance of ~~Algorithm~~ system depends as much on efficient Algo as it does on high speed Hardware. Algorithm is reffered as Technology Because:

→ They are foundational Software that enables complex tasks.

→ Even with the fastest computers a poorly designed algorithm is useless for large data making itself primary technical constraint.

4. When is an algorithm said to be correct?

Correctness:

An Algorithm is said to be correct if for every possible input instance it halts and produces the correct output then we say algorithm is Correct.

n+1

odd n → 3n+1
even n → n/2

5. The Syracuse problem is a very simple problem of arithmetic that is still unsolved today. It can be stated as follows: the Syracuse sequence starts from any natural number, the next number of the sequence depends upon its previous as, if n^{th} number is odd, the $n+1^{th}$ number is **3n+1** and if n^{th} number is even $n+1^{th}$ number is **n/2**. Write the recursive implementation of following function to print Syracuse sequence start from **n**. *Note: you may write cpp code or just algorithm.*

```
void syracuse(int n)
{
  cout << n << " ";
  if (n == 1)
      return ;
  if (n%2 != 0)  // odd
     syracause(3*n+1);
  else  //even
     syracause(n/2);
}
```

3(3)+1 = 10
10/2 = 5
3(5)+1 = 16
16/2 = 8
8/2 = 4
4/2 = 2
2/1 = 1

6. The recursive formula for 01 Knapsack problem is as follows:

$$B(n,W) = \begin{cases} 0 & \textit{when } n \le 0 \textit{ or } W \le 0 \\ B(n-1,W) & \textit{when } n > 0 \textit{ and } w[n] > W \\ \max(\ B(n-1,W), \quad B(n-1,W-w[n]) + v[n]\ ) & \textit{when } n > 0 \textit{ and } w[n] \le W \end{cases}$$

where B is the maximum benefit from **n** item and knapsack capacity **W**. The arrays **w** and **v** contain the weights and the values correspond to **n** items in the shop. Write the following recursive function to return the maximum benefit for the specified 01 Knapsack problem. *Note: you may write cpp code or just algorithm.*

```
double knapsack01(int n, int W, double w[], double v[])
{
  if( n<=0 || W<=0)
      return 0;
  if( w[n-1] > W)
      return knapsack01(n-1, W, w, v);
  else
  {
    double exclude = knapsack01(n-1, W, w, v);
    double include = V[n-1] + knapsack01(n-1, W-w[n-1], w, v);
    return max(include, exclude);
  }
}
```